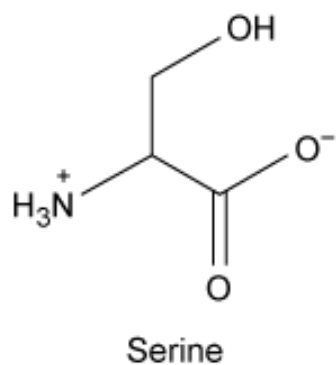
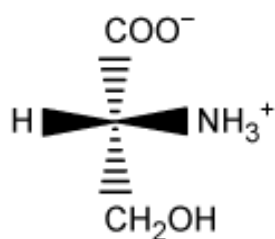


The structure of the amino acid serine is shown below.

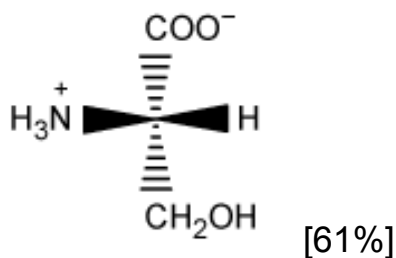


Which of the following structures of serine has an L configuration?

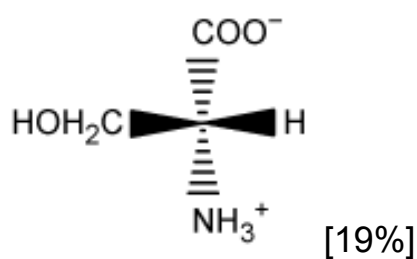
☐ A.



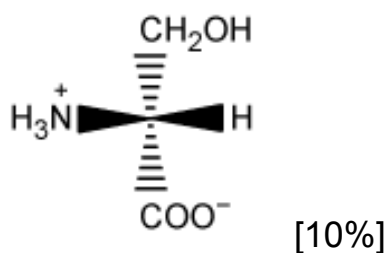
✓ ☒ B.



☐ C.



☐ D.

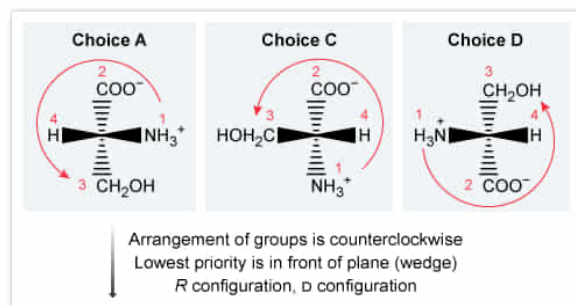
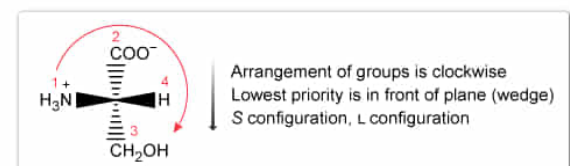


Correct

61% answered correctly

Explanation:

Configurations of serine



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Amino acids contain a central carbon bonded to an amino group, a carboxyl group, a hydrogen atom, and a side chain. The central carbon is known as the α -carbon and is a **stereocenter** in all amino acids except **glycine**, which has only three unique substituents bonded to the α -carbon (ie, two groups are hydrogen). All other amino acids contain a stereocenter because the side chain is different than the other three substituents (ie, four unique substituents are attached to the α -carbon).

These amino acids can be classified in two ways: the **L/D system** and the **R/S system**. The L/D system compares chiral molecules to glyceraldehyde whereas the R/S system assesses the arrangement of substituents around the central carbon. To determine the L/D configuration of an amino acid, it is easiest to first determine the R/S configuration by considering the **priority** of the α -carbon substituents. For most amino acids, the order of priority is

1. $-\text{NH}_3^+$
2. $-\text{COO}^-$
3. **Side chain**
4. $-\text{H}$

Cysteine is an exception because its side chain ($-\text{CH}_2\text{SH}$) has a higher priority than $-\text{COO}^-$. All **natural chiral amino acids** have an **S configuration** except for cysteine, which has an **R configuration**. However, both **R-cysteine** and the **S form** of other amino acids are in the **L configuration**; therefore, the **S form** of serine is the **L form**. From highest to lowest priority, the α -carbon substituents shown in Choice B are arranged in a clockwise fashion. Because the lowest priority group is **in front of the plane (wedge)** in the given depiction, the clockwise arrangement indicates the molecule is the **S form**, and is therefore the **L configuration of serine**.

(Choices A, C, and D) From highest to lowest priority, the α -carbon substituents in

these structures are arranged in a counterclockwise fashion with the *lowest priority* group *in front* of the plane. Therefore, these molecules are different perspectives of *R*-serine, which is the same as D-serine.

Educational objective:

Amino acids are chiral molecules (except glycine) that contain an α -carbon atom with four unique substituents: -NH_3^+ , -COO^- , a side chain, and -H . The natural chiral amino acids all have an *S* configuration except cysteine, which has an *R* configuration. All the natural chiral amino acids have an L configuration, and the enantiomers of these amino acids have a D configuration.

Time spent:0 sec

Subject:

Organic Chemistry

QID:402646

System:

Functional Groups and Their Reactions